

UR2Phd Starter Assignment Report

Setup

I built two systems for DiCo-NLI Track 1(English), which were trained on the official train split (3042 items) and scored every result on dev(660 items) with the official scorer.

Label	Train	Dev
BACKWARD ENTAILMENT	881	190
EQUIVALENCE	802	174
FORWARD ENTAILMENT	881	190
NEGATIVE OTHER	478	106

System A was fine-tuned as a 4-way sequence pair classifier: learning rate 2e-5, batch size 16, 5 epochs, seeds 13, 42, 123. The epoch count was fixed in advance, so dev was never used for model selection. System B, prompted with greedy decoding. Zero-shot uses a prompt defining the four labels; few-shot adds 8 training examples. A keyword parser maps outputs to labels; 0 of 660 were unparsed in both variants. Both systems ran on a Colab T4 GPU. Data exploration, full hyperparameters, and prompts are in the README and notebooks.

Results

Fine-tuned DeBERTa is far ahead of the prompted LLM on all three metrics, and its scores are stable across seeds.

System	Weighted F1	SoftCons	HardCons
Majority Baseline	0.129	0.000	0.000
A: DeBERTa-v3-base (mean $\pm$ std)	0.826 $\pm$ 0.014	0.868 $\pm$ 0.002	0.821 $\pm$ 0.009
B: Qwen2.5-1.5B zero-shot	0.287	0.126	0.090
B: Qwen2.5-1.5B few-shot	0.333	0.271	0.181
Pilot DeBERTa-v3-base	0.75	0.84	0.79

Runs that did not work. My first three DeBERTa runs collapsed to the majority baseline because the checkpoint loaded in float16 and the loss became NaN; converting to float32 fixed it. My first scoring of System B was wrong because of a parser bug that missed EQUIVALENCE outputs; I reran both variants with a parser checked by assertions.

Analysis

Comparisons of Systems A and B on three metrics

DeBERTa beats Qwen by about 0.5 Weighted F1 and by more on both consistency metrics. Few-shot prompting roughly doubles Qwen's SoftCons (0.126 to 0.271) and HardCons (0.090 to 0.181) but adds only 0.05 F1. DeBERTa is above the pilot's DeBERTa numbers, though the label sets differ. Qwen is well below the pilot's decoders, which were fine-tuned, while mine is only prompted. The ordering matches the pilot: encoder ahead of decoder, with the largest gap on consistency.

Labels each system confuses most

GOLD	BACKWARD	EQUIVALENCE	FORWARD	NEG_OTHER
BACKWARD (190)	165	11	2	12
EQUIVALENCE (174)	2	142	6	24
FORWARD(190)	2	8	169	11
NEG_OTHER(106)	11	12	9	74

DeBERTa rarely reverses the direction: only 4 errors swap FORWARD and BACKWARD. NEGATIVE_OTHER is involved in 79 of the 110 errors and is the weakest class (70% recall). The largest single confusion is EQUIVALENCE predicted as NEGATIVE_OTHER (24). The hard decision is whether a relation exists, not which way it points. Qwen has the opposite problem. Gold BACKWARD is correct 2 of 190 times zero-shot and 0 of 190 few-shot; it is labelled FORWARD 166 and 126 times. Qwen detects an entailment but calls it FORWARD, regardless of order. Few-shot mainly moves predictions from FORWARD to EQUIVALENCE.

10 DeBERTa errors

Pattern	Examples
Prepositions differ	“in a top hat” “with a top hat”. “onto another” “to another”. “at a pool” “in front of a pool”. “to pro-Morsi protests” “after pro Morsi protest”
Near synonyms	“eat” “having a meal”. “to deliver statement” “to speak”.
Added modifier	“game” “a video game”
Quantifier/Number	“people” “several people”. “the brownish and white dogs” “the tan and white dog”
Low word overlap	“its first memorial” “remembered”

Prepositions: when phrases differ mainly in their preposition, the model errs both ways, calling different meanings equivalent and related phrases unrelated. The pilot paper reports the same weakness. Entailment collapsed into equivalence: for close meanings, the model answers EQUIVALENCE and loses the direction. Low word overlap: a true relation in different words is labelled NEGATIVE_OTHER. Quantifiers and numbers carry the entailment and are missed.

5 pairs where your system is inconsistent under reversal

- “a video game” “game” – FORWARD then EQUIVALENCE
- “a candy bar” “a chocolate bar” – EQUIVALENCE then FORWARD
- “two Asian men and Asian women” “two Asian men and a woman” – FORWARD then EQUIVALENCE
- “fresh election for president” “fresh presidential election” – FORWARD then EQUIVALENCE
- “is skijoring” “is playing” – FORWARD then NEGATIVE_OTHER

37 of 277 pairs fail SoftCons. In 20 of 37, the model predicts NEGATIVE_OTHER in both directions; the other 17 are real contradictions. Most of these (10) are entailment one way and EQUIVALENCE the other: the model picks a direction when the specific phrase comes first and falls back to EQUIVALENCE when it comes second.

Next Step

I would predict each pair in both orders and combine the two answers, so that a pair and its reverse always get compatible labels. This should fix the 17 pairs where DeBERTa contradicts itself and raise SoftCons from about 0.87 to about 0.93, with no retraining.